

THE UNITED REPUBLIC OF TANZANIA
PRESIDENT OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DODOMA REGION FORM SIX MOCK EXAMINATIONS

CODE: 131/1

PHYSICS 1

Time; 3 Hours

November, 2023

INSTRUCTIONS

1. This paper consists of **ten (10)** questions in section A and B.
2. Answer **all** questions in section A and **two (2)** questions from section B.
3. Section **A** carries **seventy (70)** marks and section **B** carries **thirty (30)** marks.
4. Marks for each question part thereof are indicated
5. Mathematical tables and non-programmable calculators may be used.
6. Cellular phones and any unauthorized material are not allowed in the examination room.
7. Write your Examination Number on every page of your booklet(s).
8. The following information may be useful;-
 - (i) Acceleration due to gravity, $g = 10\text{ms}^{-2}$
 - (ii) Molar gas constant $R = 8.31\text{ Jmol}^{-1}\text{K}^{-1}$
 - (iii) Gravitational constant $G = 6.67 \times 10^{-11}\text{Nm}^2\text{kg}^{-2}$
 - (iv) Radius of the earth $R_E = 6.4 \times 10^6\text{m}$
 - (v) Thermal conductivity of X $= 400\text{Wm}^{-1}\text{K}^{-1}$
 - (vi) Thermal conductivity of Y $= 200\text{Wm}^{-1}\text{K}^{-1}$
 - (vii) Ratio of heat capacities for air $\gamma = 1.4$
 - (viii) Electronic charge $e = -1.6 \times 10^{-19}\text{C}$
 - (ix) Pie $\pi = 3.14$

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SECTION A (70 marks)

1. (a) (i) Is it possible for a physical quantity to be dimensionless? Give one example to support your answer. (2 marks)
 (ii) Give two physical quantities which have the same dimension as that of work. (2 marks)
- (b) (i) Applying a formula where it does not hold, is it a mistake or an error? Briefly explain by giving the difference between the two terms. (2 mark)
 (ii) Calculate the percentage error of the quantity $Z = xy - b^2$, where $x = (3.00 \pm 0.15)cm$, $y = (5.00 \pm 0.17)cm$ and $b = (3.00 \pm 0.13)cm$ (4 marks)
2. (a) (i) State the principle of conservation of linear momentum and show how it follows from Newton's laws of motion. (3 marks)
 (ii) Sand falls at the rate of 0.15kg s^{-1} on to a conveyor belt moving horizontally at a constant speed of 2m s^{-1} . Calculate the extra force necessary to maintain this speed. (2 marks)
- (b) A ball is thrown horizontally with a speed of 30m/s from the top of building at height 50m . Determine the time of flight and maximum horizontal distance reached by the ball. (5 marks)
3. (a) A steel strip clamped at one end, vibrates with a frequency of 20Hz and amplitude of 5mm at the free end, where a 2g mass is positioned. Find:-
 (i) the velocity of the mass when passing through zero position. (2 marks)
 (ii) the acceleration at maximum displacement. (2 marks)
 (iii) the maximum kinetic and potential energy of the mass (2 marks)
- (b) A sphere of mass 0.5kg is tied to one end of the string 1m long, the point of suspension of the string is 2m above the ground. The stone is whirled in the horizontal circle with increasing angular velocity and the string breaks when the tension in it is 12.5N and the maximum angle θ_{max} to the vertical as shown in figure 1.

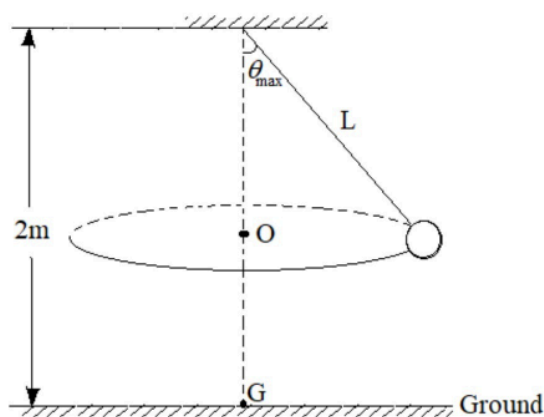


Figure 1

- (i) Calculate the angular velocity and maximum angle when the string breaks (2 marks)
- (ii) How far from point G on the ground will stone hit the ground? (2 marks)

4. (a) (i) Explain what is meant by the universal gravitational constant G . **(2 marks)**
(ii) If two uniform spheres each of mass M and radius R , touch one another, show that the magnitude of the gravitational force is $\frac{G}{4} \left(\frac{M}{R} \right)^2$ **(4 marks)**
- (b) A solid ball and a solid cylinder roll down a slope. Show that the sphere will reach the bottom first. **(4 marks)**
5. (a) (i) Explain how centigrade temperature scale is defined. Illustrate your answer by reference to a platinum resistance thermometer **(1 mark)**
(ii) The resistance R_t of a platinum wire at a temperature t °C is given by $R_t = R_o (1 + at + bt^2)$ where $a = 3.800 \times 10^{-3}$ and $b = -5.6 \times 10^{-7}$. What temperature will platinum thermometer indicate when the temperature on the gas scale is 200°C? **(4 marks)**
- (b) (i) All objects radiates electromagnetic energy. Why do they not lose all their thermal energy by radiation and cool to 0K? **(1 mark)**
(ii) Two perfect lagged bars X and Y are arranged in series and parallel. When the bars are in series the hot end X is maintained at 90°C and the cold end of Y is maintained at 30°C. When the bar are in parallel the hot end of each is maintained at 90°C and the cold end is maintained at 30°C. Calculate the ratio of the total flow heat in parallel arrangement to that of series arrangement. The length and cross-sectional area of each bar is the same. **(4 marks)**
6. (a) (i) Explain applications of first law of thermodynamics in our daily life. Give three points **(3 marks)**
(ii) What amount of heat must be supplied to 20g of nitrogen at room temperature to raise its temperature by 50°C at constant pressure? **(3 marks)**
- (b) A motor car tyre has a pressure of four atmospheres at a room temperature of 30°C. If the tyre burst, calculate the temperature of escaping air. **(4 marks)**
7. (a) (i) Briefly explain the formation of an Earthquake. **(3 marks)**
(ii) Briefly explain three techniques used to improve plant environment. **(3 marks)**
- (b) A solar cell of surface area 100cm² is illuminated by monochromatic light of a wavelength of 760nm and power density of 1000Wm⁻². The open circuit voltage is 0.67V when the cell temperature is 300K. Calculate the dark cell current I_o , assuming a 100% quantum efficiency **(4 marks)**

SECTION B (30 marks)

8. (a) (i) Explain the rules which are useful in solving circuit problems. **(2 marks)**
(ii) Study the figure 1, calculate the current through ammeter A_1 , ammeter A_2 and the potential difference across R_2 . Assume the internal resistance of each cell and resistance of each ammeter is negligible. **(3 marks)**

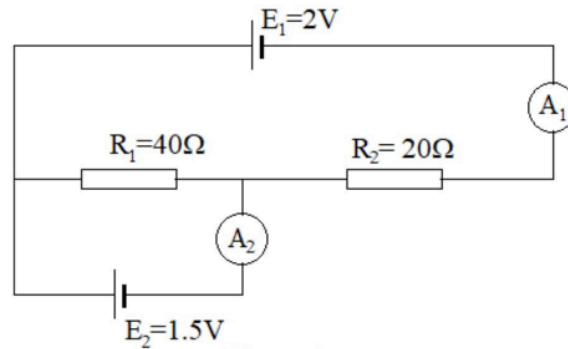


Figure 2

- (b) A power supply used in a battery has an e.m.f of 5000V. When a voltmeter of resistance $20\text{k}\Omega$ is connected to the terminal of the power supply a reading of 40V is obtained.
- Explain this observation. **(2 marks)**
 - Calculate the current flowing through the meter and the internal resistance of the power supply **(3 marks)**
- (c) (i) Distinguish impedance from the power factor. Give one point. **(1 mark)**
- (ii) Consider an LCR series circuit with $R = 300\Omega$, $L = 0.9\text{H}$ and $C = 2.0\mu\text{F}$ where the source frequency is 50Hz and the voltage is 240V . Determine; the impedance and the current through the circuit, the phase angle and the voltage across each component. **(4 marks)**

9. (a) (i) Explain the significance of Fermi level and energy gap in solids. **(2 marks)**
- (ii) Analyse the effect of temperature on electric conduction of solid. **(2 marks)**
- (iii) Use the band theory of solids, explain why insulators do not conduct electric current. **(1 mark)**
- (b) (i) With the aid of diagrams, explain the forward bias and reverse bias mode of p-n junction hence justify that, p-n junction is also called a semiconductor device. **(4 marks)**
- (ii) Silicon is preferred to germanium in the manufacturing of semiconductor's devices. Give reason to support this statement. **(1 mark)**
- (c) Study the figure 3, then answer the questions than follows

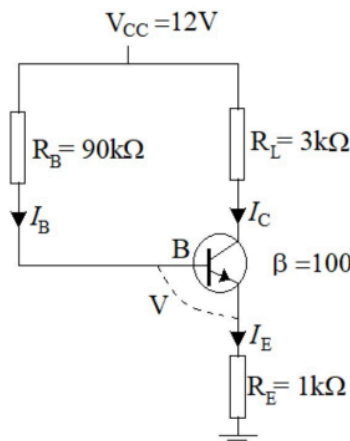


Figure 3

- (i) Show that silicon transistor will operate in saturated region. **(1 mark)**
- (ii) Calculate the value of R_E so that the transistor is just out of saturation. **(4 marks)**

10. (a) Use the Boolean algebra rules to prove that;-

- (i) $A + A \cdot B = A$ **(1 mark)**
- (ii) $A + \bar{A} \cdot B = A + B$ **(2 marks)**
- (iii) $(A + B) \cdot (A + C) = A + B \cdot C$ **(2 marks)**

(b) Give the meaning of the term “logic gate”, hence briefly explain three basic logic gates. **(3 marks)**

(c) Study the figure 4, then answer the questions that follows;-

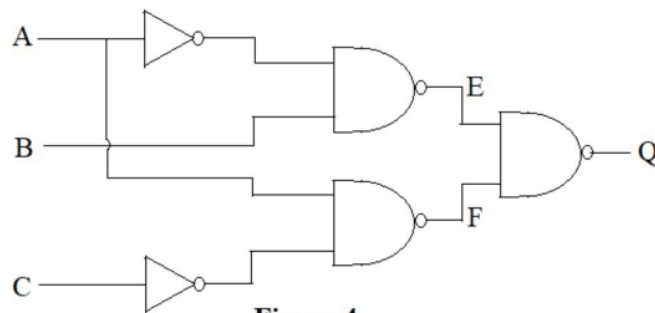


Figure 4

- (i) What type of a single gate would provide exactly the same function as a circuit in a figure 4. **(2 marks)**
- (ii) Draw the symbol of the logic gate representing a single gate mentioned in (c) (i) **(3 marks)**